The Calendars - Cal.diy

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Sprint 2 Overview

In this sprint, we continued adding new features. We cleared up some long-standing issues such as the two factor authentication and waitlists, finished up some new issues spawned from Sprint 1 such as the availability calendar dragging, and added new features like the recurrence fields.

We were able to complete **9** out of 13, which was around **56** out of 74 total story points. This is in part due to dchan01-wm's issues, where he discovered that they were significantly larger issues than anticipated.

UML Diagrams

- **Link to the diagrams:**
  - cal.diy/docs/sprint2/diagrams at main · CSCI-435-SE/cal.diy

Mid-Sprint Checkins

- **Link to the checkins:**
  - cal.diy/docs/sprint2/checkin at main · CSCI-435-SE/cal.diy

Sprint 1 Holdovers

Issue #	Issue name	Status	Reason was Heldover/Removed
#19	Allow users to set a custom default landing page after login	Done in Sprint 1	Accidentally forgot to close the issue
#51	Add a rebook feature	Done in Sprint 1	Parent issue of two completed subissues; accidentally forgot to close the parent
#50	Add a calendar view for bookings	Moved to **Sprint 2**	N/A - carried over
#52	Add bulk actions to the bookings list	Moved to **Sprint 2**	N/A - carried over

Sprint 2 Backlog

- **Link to the milestone:**
  - Milestone 2 - https://github.com/CSCI-435-SE/cal.diy/milestone/2

Issue #	Issue name	Assignee	Done?	Est. story points	Rationale
#5	Ability to set event theme	`zach-short` 	No: PR #122 open	5	It adds stored colors per event type, picker UI in the event type editor and conditional styling on the booking page, across about 6–8 files. The logic is simple (hex validation, contrast check) and follows the existing `eventTypeColor` pattern, but the light/dark theme interaction adds some design work.
#13	Add a waitlist for fully-booked time slots	`zachbowden` 	Yes: PR #125 merged	13	It needs a new Prisma model and migration, hooks into the cancel and reschedule paths, notifications, host settings UI and attendee UI. Time-limited claim links need race handling between claimants and must respect `seatsPerTimeSlot`, so it is XL and should be split.
#50	Add a calendar view for bookings	`dbbeynam`	No: PR #95 open	5	It is frontend-only: a new month-grid component, a three-way view toggle and URL/saved-view precedence, with no schema or API changes. Placing bookings on the right day across timezones and DST and loading a full month of bookings need careful logic and tests.

#52	Add bulk actions to the bookings list	`dbbeynam`	No: PR #98 open	5	It reuses the existing cancel endpoint but adds a checkbox column with eligibility rules, selection state, a selection bar and a confirmation dialog. The batch logic needs sequential requests, a 10-item cap, partial-failure reporting and "retry failed only".
#57	Grace period to undo a cancellation	`zachbowden` 	Yes: PR #114 merged	8	It needs a new soft-cancel state, an undo toast and a background job that finalizes the cancellation after the grace period. Every cancellation side effect (emails, calendar deletion, webhooks, refunds) has to be delayed, which restructures a critical path.
#63	Day-of-Week Recurrence Selection	`dchan01-wm` 	No: **split** into #117 and #118	8	It adds weekday selection to the `recurringEvent` data, a day-toggle row in the recurrence UI and changes to how occurrences are generated and booked. The existing RRule function does not cover all cases; more complicated than expected.
#64	Configurable Recurrence End Conditions	`dchan01-wm` 	No: **pushed to Sprint 3**	5	It adds an "Ends" section (Never / On date / After N times) and an `until` field, which RRule already supports. "Never" is harder than it looks because each occurrence is a booking row, and the issue is light on acceptance criteria. **In the process of implementing Issue 63**, we realized that this would be better off in Sprint 3, so dchan01-wm could focus on #63.

#71	Add Toggle to Switch Between Text-Field and Calendar Grid Modes (Subissue of #32)	julissaehp 	Yes: PR #116 merged	2	It is a visibility toggle in one component (`AvailabilitySettings.tsx`) with session persistence and a few i18n strings. Data consistency between the two modes is left to #72, which keeps the logic minimal.
#93	Add long-press to start drag on touch devices in the weekly availability grid	julissaehp 	Yes: PR #115 merged	5	The change is limited to about 3 files (`useWeeklyGridDrag.ts`, the grid component and tests), but pointer-event handling is fiddly. It needs a hold timer cancelled by movement or `pointercancel`, a non-passive `touchmove` listener and callout suppression, plus manual device testing because jsdom can't simulate scrolling.
#104	Show the away badge and redirect for partial-day out-of-office entries	zach-short 	Yes: PR #110 merged	8	It touches the core availability engine (`getUserAvailability.ts`, `date-ranges.ts`, `slots.ts`) and changes the out-of-office data shape to carry time windows. The booker also has to render the away panel mid-day, with DST and booking-blocking edge cases, which puts the change near the PR size limit.
#111	Add an admin setting to require 2FA for all users	kojoshuay 	Yes: PR #113 merged	3	It stores one site-wide setting with an admin-only toggle and server-side admin checks, following existing admin settings patterns. The logic is simple, and most of the effort goes into the access-control tests.

#112	Require users to set up 2FA when the admin setting is on	`kojoshuay` 	Yes: PR #120 merged	5	Regular users have no existing enforcement, so it needs a session flag set at login, an app-wide setup modal that can't be closed and a required onboarding step. It crosses the auth/session layer, the shell UI and onboarding, and lockout risk and SSO exemptions make careful testing necessary.
#117	Choose which weekdays a weekly recurring event repeats on	`dchan01-wm` 	Yes: PR #117 merged	7	It is the host side of #63: day buttons on the Recurring tab, saving the days with the host's time zone and week start, and generating the series with RRule `BYDAY`/`WKST`. It also adds server-side rejection of a first slot on an unselected day, which touches the recurring booking service and its tests. Unexpectedly more complicated than anticipated. #118, the booker counterpart of this issue, was **left out of Milestone 2** due to complexity with #117.

Requirements and Design

kojoshuay

- The primary design dilemma was determining the most efficient database architecture to store a single site-wide security boolean without introducing unnecessary technical debt.
    - Adding a new boolean column to the Deployment table was rejected because it would require building new repository methods, endpoints, and admin checks from scratch.
    - Creating a SiteSettings table was rejected as over-engineering since a single boolean setting does not justify building a new schema, repository, and DI container.
    - Storing the setting as a row in the existing Feature table was chosen because it requires zero schema changes and safely reuses the application's established backend architecture for site-wide rules.

zachbowden

- For the soft cancellation feature, we chose a five-minute window before a cancellation actually goes through. That gives the user enough time to reconsider without leaving the

host uncertain for long. During that window, the host's panel still shows the booking as active, and it disappears only once the five minutes are up. For the waitlist feature, we chose to give a user six hours to accept an offered spot. That is enough time for them to work out whether they can attend, and for the host to adjust their plans for the meeting. We also decided to support both whole-day and specific-time waitlists, each used in different situations. The whole-day option appears only when every time slot that day is booked, and on any other day users see a waitlist option for each booked time.

Completed Issues

Issue	Owner	PR	PR author	Reviewer(s)	PR status	Description of changes
#71 Add Toggle to Switch Between Text-Field and Calendar Grid Modes	`julissaehp`	#116	`julissaehp`	`kojoshuay`	Merged	Adds a Text / Visual toggle to the availability settings page so one editor (the existing text fields or the visual grid) shows at a time. The choice is saved in `sessionStorage` for the tab, the toggle supports the keyboard, and 15 tests were added. Syncing data between the two modes is left to #72. (+339/−1, 8 files)
#93 Add long-press to start drag on touch devices in the weekly availability grid	`julissaehp`	#115	`julissaehp`	`zachbowden`	Merged	On touch, a drag now starts only after a 350 ms hold, and moving more than 8 px first cancels it, so a quick swipe scrolls instead of selecting. Once a drag starts, scrolling and the context menu are blocked, and a second finger cancels the drag. Mouse and pen behavior is unchanged, and tests were updated. (+339/−36, 4 files)

″	″	#121 (docs)	`julissaehp`	`zachbowden`	Merged	UML diagrams and Q&A for the #93 design. (+49, 3 files)
#104 Show the away badge and redirect for partial-day out-of-office entries	Zachary Short (`zach-short`)	#110	`zach-short`	`dchan01-wm`	Merged	The per-day out-of-office map now carries partial-day time windows, so slots that overlap the window are marked `away` and still can't be booked. The booker shows one away panel (reason, emoji, note, "Book with" redirect) between the slots before and after the window. Includes DST, multi-window and booking-guard tests. (+839/−117, 10 files, 661 of the lines are tests)
″	″	#123 (docs)	`zach-short`	`julissaehp`	Merged	Sprint 2 check-in and the #104 design diagrams. (+50,723, 13 files, mostly diagram assets)

#111 Add an admin setting to require 2FA for all users	kojoshuay	#113	kojoshuay	zachbowden	Merged	Seeds a `require-two-factor-auth` row in the `Feature` table, with no schema change. Adds a `/settings/admin/security` page with an admin-only toggle and a confirmation dialog, and a live, uncached read helper (`isTwoFactorRequiredSiteWide`). The flag is hidden from the generic feature flag list. Tests cover the helper, the UI and the tRPC admin-only checks. (+7,848/−17, 18 files)
#112 Require users to set up 2FA when the admin setting is on	kojoshuay	#120	kojoshuay	julissaehp	Merged	Adds a `requiresTwoFactorSetup` flag that the server recalculates on every session read, so the browser can't skip it. When the flag is set, an un-closeable 2FA setup modal is mounted in `Shell.tsx`. New password users get a required 2FA onboarding step behind a server-side route guard, and SSO users are exempt. (+15,746/−19, 22 files)
″	″	#124 (docs)	julissaehp	dchan01-wm	Merged	AI log of the review of PR #120. (+7,762, 1 file)

Totals: 5 completed issues, 8 associated PRs (5 feature PRs and 3 docs PRs), all merged.

Test strategy

kojoshuay

- Added component level unit tests to verify:
    - 2FA popup closes upon cross tab completion but stays open while displaying backup codes
    - An “already enabled” server response completes the setup instead of throwing an error
- These tests mock sessions and server responses, so they don’t validate real NextAuth cross tab synching or actual browser behavior. Still needs manual verification.

zachbowden

- Waitlist
    - Unit tests on WaitlistService (12 tests)
    - In-memory fake database, fixed clock, mocked email and scheduler
    - Covers joining, offering, expiry, leaving, claim links and booked times
    - Includes the 6-hour cutoff, timezones, duplicates and first-come order
    - DatePicker tests updated to mock the booked-times hook
- Cancellation undo
    - Unit tests on the delayed cancelBooking task (2 tests)
    - Real cancellation handler mocked out
    - Checks the task cancels with the right data and user
    - Checks a bad payload is rejected and nothing is cancelled

dchan01-wm

- Two main strategies:
    - Unit tests:
        - Specific logic within the Recurring events, for example how the code uses rrule to generate a list of dates
    - Manual testing:
        - Mainly for UI elements where it is hard to assert programmatically how the UI looks or reacts. Used with the recurring events

AI Tool Usage

kojoshuay - I used Claude code to help implement my two issues concerning 2FA.

dchan01-wm: I used Claude Code to help review and implement issues. I used Claude to generate skills which I later used to help me review and implement.

Julissaehp - I used Claude AI to help me understand how to approach implementing Issue #67 and navigate the Cal.diy codebase. Since PR #120 had already implemented mandatory two-factor authentication, Claude helped me identify which existing files, components, and middleware I should look at instead of building the feature from scratch. It also helped me understand how the authentication process works, how administrator permissions are checked, and where the new admin-only 2FA requirement would need to be added.

Claude also helped me compare different implementation approaches, including using server-side authorization checks instead of relying only on a popup, reusing the existing 2FA components, and creating a separate admin-only setting. This helped me organize my acceptance criteria, design decisions, and implementation plan. Rather than just following AI suggestions, I used them to better understand the codebase and make decisions about how to implement the feature while avoiding unnecessary changes to existing functionality. https://github.com/CSCI-435-SE/cal.diy/pull/119/changes

zachbowden - I used Claude code to help me implement my two significantly more difficult pull requests this time.

Release

- Tag Name: v6.2.0-csci435-s2
- Link: https://github.com/CSCI-435-SE/cal.diy/releases/tag/v6.2.0-csci435-s2

Risks and Retrospective

What went well? What slowed you down? What would you do differently in Sprint 3?

- More aware of the code architecture this time, so more accurately able to assign story points; although still had some problems with estimating story points (dchan01-wm)
- Attempted more automation, but the set-up took a long time
- Can lean on the automation in Sprint 3

Sprint 3 plan

Continue work on the backlog of issues. Since we seemingly have a team mate who is missing, we need to adjust our workload accordingly.